

Google open sources PSP Security Protocol



Google has open sourced an encryption scheme it had developed to protect traffic between its data centres. PSP, which stands for PSP Security Protocol, was created to relieve Google's processors of the growing burden of software-based encryption, according to the company. PSP has been hailed as a success in the company's own environment, and the company has stated that it is "making PSP open source to encourage broader adoption by the community and hardware implementation by additional NIC [network interface card] vendors." PSP offloads encryption to NICs, which was previously possible with existing encryption schemes, but not at the scale or with the traffic coverage required by Google.

"At Google's scale," the company wrote when announcing its decision, "the cryptographic offload must support millions of live transmission control protocol (TCP) connections and sustain 100,000 new connections per second at peak."

Existing security protocols, according to Google Cloud's Amin Vahdat and Soheil Hassas Yeganeh, had flaws. "While TLS meets our security requirements, it is not an offload-friendly solution because of the tight coupling between the connection state in the kernel and the offload state in hardware. TLS also does not support non-TCP transport protocols, such as UDP," they stated.

Google added a custom header and trailer to the standard User Datagram Protocol (UDP) encapsulation to create PSP. PSP is currently implemented in three ways: one for Google's Andromeda Linux virtualisation kernel; one for its Snap networking system; and an application-layer version, SoftPSP, created so Google Cloud customers could use PSP on computers with traditional NICs.

Iterative releases MLEM, an open source machine learning tool

Iterative, an MLOps platform that raised US\$ 20 million in a Series A round almost a year ago, has launched MLEM, an open source Git-based machine learning model management and deployment tool. The goal here, according to the company, is to bridge the gap between ML engineers and DevOps teams by utilising the Git-based approach that developers are already familiar with. Developers can use MLEM to store and track their machine learning models throughout their life cycle. As such, it

Kubecost develops OpenCost to rein in K8s costs

Kubecost's tool for monitoring and optimising spending on Kubernetes clusters has been released as an open source project. OpenCost is now available as open source software, according to Alex Thilen, head of business development at Kubecost, and the company has already submitted it to the Cloud Native Computing Foundation (CNCF) for approval as a sandbox-level project. Adobe, Armory, Amazon Web Services (AWS), D2iQ, Google, Mincurv, New Relic, and SUSE are among the project's founding members, in addition to Kubecost.

Although OpenCost is designed to run within a Kubernetes cluster, no data is sent outside of the cluster without user permission. It can collect data in real-time after only a few minutes of installation.

The primary issue addressed by OpenCost is overprovisioning of Kubernetes infrastructure. Many developers overprovision infrastructure to ensure maximum application performance. The problem is that much of that infrastructure is unused, and costs rise steadily as each new Kubernetes cluster is provisioned. According to Kubecost, organisations can cut Kubernetes-related cloud spending by 60-80 per cent without sacrificing application performance.

As the percentage of workloads running on Kubernetes clusters grows, it is more likely that those platforms will be managed centrally by an IT operations team. These teams are graded based on how well they optimise cloud infrastructure usage. They must also demonstrate to development teams how much Kubernetes infrastructure is consumed by individual applications.